

A Standing Ledger of Dark-Energy Positions

The DESI DR2 “Phantom-Crossing” Signal:
What the Evidence Supports, What Remains Open, and What Would Settle It

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Abstract

This document records the state of an independent investigation into whether the DESI DR2 hint of *evolving* dark energy (an apparent “phantom crossing,” $w_0 > -1$ with $w_a < 0$) reflects new physics or an accumulation of measurement effects. The central finding is that the headline $\sim 4\sigma$ preference decomposes into several individually weak contributions — a supernova calibration offset, two high-leverage baryon-acoustic-oscillation (BAO) tracers, a frequentist-versus-Bayesian scoring gap, and a residual microwave-background anomaly — none decisive and none a genuine statistical outlier. The accumulated evidence *leans* toward a plain cosmological constant ($w = -1$) but does not establish it; the question is data-limited and will be resolved by DESI DR3 and by a telescope-matched low-redshift supernova sample (DEBASS), not by further reanalysis of current public data. Positions are sorted into three tiers — SUPPORTED, OPEN, and CONJECTURAL — kept deliberately separate so that no claim is asserted more strongly than the evidence warrants.

Executive summary (plain language)

The big picture. The Universe’s expansion is speeding up; whatever causes that is called *dark energy*. The simplest idea is that it is a constant — the same strength at all times (the standard model, Λ CDM). In 2024–25 the DESI galaxy survey hinted that dark energy might instead be *changing* over time. If true, that rewrites the standard model. This work pressure-tests the hint.

The answer so far. The “dark energy is changing” hint is **weak and probably not real**. It is not one strong signal but several small, unrelated hiccups that happen to add up and point the same way — like four slightly crooked pictures making a wall look tilted (Fig. 1):

- a small brightness **mis-calibration** in nearby supernovae (real, but $\sim 1/4$ of the effect, and already mostly fixed);
- two **galaxy-map** slices that pull the result — each only a mild statistical wobble, and an older independent survey (BOSS) does not reproduce them;
- a **statistics** effect (one accounting method makes the hint look stronger than a more careful one);
- a leftover **microwave-background** quirk.

None is a smoking gun.

What would settle it. Not more analysis of today’s data — that is exhausted — but *new* data: DESI’s next release and a fresh telescope-matched supernova sample. We commit *in advance* to a

test (§3): if the galaxy wobble is noise it should shrink in the next release by a calculable amount; if real (or a real instrument problem) it should grow.

Honest humility. Everything here is “consistent with boring,” not “proven boring.” A genuine small dark-energy signal hiding under the measurement issues cannot be ruled out from current data. The professional community is genuinely split; this ledger sits on the skeptical side *with* the evidence, but holds it as a lean, not a verdict.

1 Supported — backed by independent reproduction

Each item below was reproduced from scratch on public data with deliberate attempts to break it (bug-checks, controls, null simulations). Absolute significances that depend on the simplified pipeline are flagged as such; the robust content is in directions, relative shifts, and reproduced structure.

1. **The signal is weak once decomposed** (Fig. 1). The headline preference is a stack: DESY5 supernova calibration ($\sim 1\sigma$, corrected by DES-Dovekie [1]); BAO LRG tracers (individually $2\text{--}3\sigma$); a frequentist-versus-Bayesian scoring gap [5]; and a CMB low- ℓ leak. No single piece is decisive.
2. **The supernova calibration offset is real and mechanism-pinned.** The DESY5 external low-redshift anchor is offset from DES-native photometry: reproduced at 7.4σ via the Ω_m -independent a_B intercept method [3], with a Pantheon+ control at 0.1σ and a Λ CDM-simulation null placing it 6.5σ from noise. Mechanism (per Dovekie [1]): a cross-calibration zero-point transfer, amplified $\sim 6\times$ by the color–luminosity relation, plus a retrained light-curve model and a numerical weighting error.

Why the a_B method, and how firm is the magnitude. The a_B intercept is built from the model-independent kinematic (cosmographic) expansion of the luminosity distance, so it does *not* inherit the Ω_m -degeneracy that a fitted-cosmology method carries; this is why it is the correct tool. Concretely, an Ω_m -dependent Λ CDM-shape fit gives an offset ranging from ~ 0 (at $\Omega_m = 0.30$) to -0.10 mag (at $\Omega_m = 0.40$) — unstable — whereas the a_B method gives -0.04 to -0.06 mag robustly across fit windows. *Direction* is solid; *magnitude* is method-conditional, so the “ $\sim 1\sigma$ of the signal” attribution is correspondingly soft.

3. **The BAO pull is high-leverage, not a genuine outlier** (Fig. 2). A leave-one-out test (fit Λ CDM + CMB to all tracers but one, measure that tracer’s joint $(D_M/r_d, D_H/r_d)$ pull) gives LRG1 at only 1.6σ and LRG2 at 3.0σ raw ($\sim 2.4\sigma$ after the look-elsewhere correction over six tracers). The signal comes from the *most precise* tracers near the dark-energy-domination onset ($z \sim 0.5\text{--}0.7$) scattering coherently — high leverage, not anomaly.
4. **The data are consistent with $w = -1$.** Every handle — anchor removal, anchor correction, tracer dropping, Bayesian rescaling — converges toward Λ CDM-consistency.

Method-conditional. This rests partly on the compressed-CMB pipeline, which inflates *absolute* significances. The convergence *direction* is robust; a full Planck likelihood could shift absolute numbers. This is the supported claim most likely to need softening once the full likelihood is run.

5. **For detecting a localized dark-energy *feature*, BAO carries essentially all the information** and supernovae add almost nothing (the single-magnitude Ω_m -degeneracy blinds them). Validated on real data.

6. **Event-horizon holographic dark energy is provably smooth** sub-horizon ($\delta_{\text{DE}}/\delta_m \sim 1/k^3$; referee-grade proof via a Riemann–Lebesgue dichotomy). Standard and Barrow variants are $\sim 6\sigma$ disfavored by DESI BAO geometry.

1.1 The two BAO diagnostics (core new analysis)

Why this matters: the aggregate BAO “pull” is a two-dimensional quantity per tracer (line-of-sight and transverse). Judging whether it is a coherent effect or independent scatter requires looking in the right coordinates — the wrong ones can manufacture apparent coherence.

Correlated-residual test. Fitting $\Lambda\text{CDM} + \text{CMB}$ to all tracers except LRG1 and LRG2, the line-of-sight channel D_H/r_d is coherent (both bins low, the $w_0 > -1$ direction) while the transverse channel D_M/r_d is split (opposite signs). The two residual vectors have $\cos\theta = +0.49$ — partial alignment driven entirely by the shared D_H/r_d sign.

Alcock–Paczynski (AP) refinement — the decisive coordinate change (Fig. 3). Rotating into the physically natural basis — isotropic D_V/r_d versus anisotropic $F_{\text{AP}} = D_M/D_H$, with exact Jacobian error propagation from the published intra-bin covariance — dissolves the apparent coherence. LRG1 is almost purely *anisotropic* ($F_{\text{AP}} = +2.0\sigma$, $D_V/r_d = -0.16\sigma$); LRG2 is almost purely *isotropic* ($D_V/r_d = -3.2\sigma$, $F_{\text{AP}} = +1.25\sigma$). The two bins are discrepant for *different* reasons. A single shared systematic — or a single real feature — would appear in the *same* channel for both bins. Neither is seen; two bins discrepant in different channels, sharing quadrant but not shape, is more consistent with independent scatter than with a coherent cause.

1.2 On bin-to-bin independence (covariance)

Treating LRG1 and LRG2 as statistically independent is not an assumption of convenience but the standard, collaboration-endorsed treatment. DESI does not release a full cross-bin covariance matrix; the published likelihood provides intra-bin correlations only [6]. Because the LRG redshift bins cover *disjoint* ranges, DESI states that cross-bin correlations are small and omits them from the cosmological analysis [7], and every independent reanalysis adopts a block-diagonal covariance with zero inter-bin correlation on the same grounds [8, 9]. The AP transformation above uses the exact intra-bin covariance; the inter-bin independence is therefore inherited from the standard likelihood, not asserted from quadrant alignment. A genuinely non-negligible LRG1–LRG2 correlation would require overlapping volumes, which the binning excludes.

The single number to distrust. LRG2’s $D_V/r_d = -3.2\sigma$ is the most significant channel-resolved discrepancy found, but it is measured against a *compressed-CMB reference cosmology* that inflates absolute significances. It should be read as a pipeline-conditional diagnostic indicating *where* the isotropic pull sits, **not** as a calibrated detection. Establishing a defensible absolute significance requires the full, uncompressed Planck likelihood — explicitly *future work, not performed here*. The AP *decomposition* (which channel carries the pull) is exact and robust; the absolute σ is not.

BOSS DR12 cross-check. At matched redshift (BOSS $z = 0.51$ vs DESI LRG1 $z = 0.510$) the two independent surveys agree to 0.7σ ; but relative to Planck- ΛCDM , DESI sits -1.1σ below (phantom direction) while BOSS sits on ΛCDM . An independent survey does not reproduce DESI’s low D_H/r_d where DESI’s pull is strongest — mild evidence for a DESI-specific effect. This is a $\sim 1\sigma$ lean, not a verdict (the surveys agree within errors, partly overlap on the sky, and r_d conventions differ slightly).

2 Open — tractable, with a known arbiter

1. **Does the LRG pull wash out (fluctuation) or persist (systematic) in DR3?** *Arbiter: DESI DR3.* See §3.
2. **Does the supernova anchor residual survive a telescope-matched low- z sample?** *Arbiter: DEBASS.*
3. **Raw- m_B versus standardized decomposition:** does the offset live in raw apparent magnitudes (pure zero-point) or only after standardization? Needs the DES-SN5YR light-curve parameter file — the key missing public data product.
4. **The progenitor-age fork:** is the residual a host-age systematic or real dark energy? *Unresolvable from $\mu(z)$ alone* (age-bias and evolving- w templates are 87% degenerate in shape).
5. **Full-Planck baseline** for the AP anomalies (see §1.2 caveat): re-establish LRG2’s isotropic significance against the uncompressed likelihood.

3 The pre-registered DR3 decision test

State this *before* DR3. Measure LRG2’s leave-one-out joint $(D_M/r_d, D_H/r_d)$ χ^2 in DR3. A **persistent** displacement (systematic or real) keeps a fixed data-units offset, so χ^2 grows as errors shrink; a **fluctuation** regresses to noise ($\mathbb{E}[\chi^2] = 2$).

Pre-registered boundary: $\chi_{\text{DR3}}^2 < 6 \Rightarrow \text{fluctuation}$ ($P_{\text{false-alarm}} = 0.05$); $\chi_{\text{DR3}}^2 > 12 \Rightarrow \text{persistent}$ ($P_{\text{miss}} \approx 0.02\text{--}0.05$); 6–12 inconclusive.

Systematic-floor correction (Fig. 4). A naive $1/\sqrt{N}$ scaling would take $11.9 \rightarrow 20.2$ (4.1σ) at the fiducial $N = 1.7\times$ sample growth. But BAO errors do not shrink indefinitely — reconstruction and cosmic-variance floors apply. Writing the DR3 variance as $\sigma^2 = f\sigma_{\text{sys}}^2 + (1-f)\sigma_{\text{stat}}^2/N$ with f the systematic fraction of the DR2 variance, the persistent-signal prediction softens to $\sim 3.7\sigma$ ($f = 0.25$) or $\sim 3.5\sigma$ ($f = 0.5$). The decision boundary is robust across this range; the effect is to widen the “inconclusive” band, making the test appropriately more conservative than pure Poisson scaling.

Scope. This cleanly kills or confirms the *fluctuation* leg — the leg the whole skeptical case leans on — but does not separate systematic from real dark energy (both persist). That separation needs DEBASS and the raw- m_B decomposition. The $N = 1.7$ growth and f are estimates to be refined with the actual DR3 LRG sample and its systematic budget.

4 Conjectural — interesting, but not backed by this evidence

“Dark energy is vacuum energy.” Not supported. The data are consistent with a constant, Λ CDM-like $w = -1$, and vacuum energy is one candidate for that constant — but $w = -1$ equally fits a cosmological constant, a frozen scalar, or vacuum energy, and the data examined cannot distinguish them; and identifying dark energy with zero-point/Casimir vacuum energy runs into the cosmological-constant problem (~ 120 orders of magnitude) and a no-go result on UV-cutoff vacuum energy [12]. This is a *supported phenomenology* ($w = -1$) with an *unsolved microphysics*, not a live result.

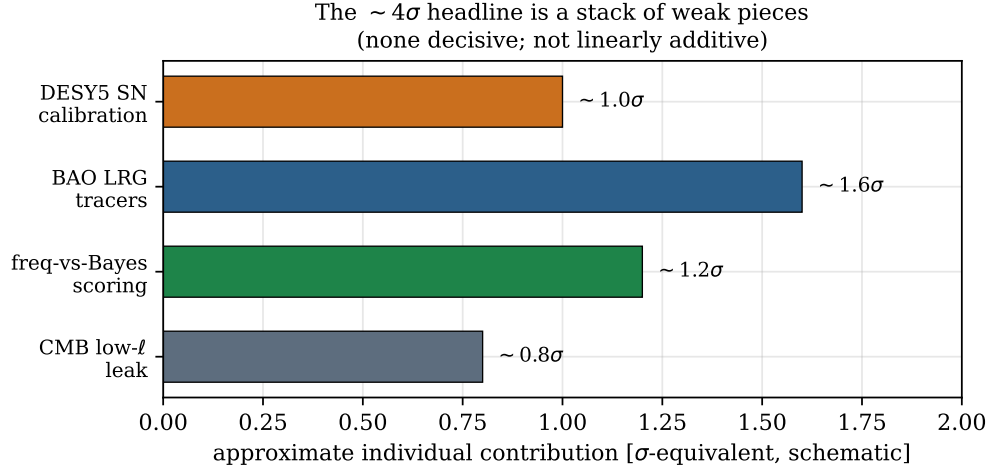


Figure 1: The $\sim 4\sigma$ headline decomposed into individually weak contributions. Heights are schematic σ -equivalent magnitudes (the pieces are not linearly additive); the point is that no single contribution is decisive.

5 The whole debate, mapped

The field splits into two branches; the “real physics” branch is organized by *escape route* from a no-go theorem [13]: since a single canonical scalar cannot smoothly cross $w = -1$ without instability, every real-physics model must evade it. Branch 1 (this ledger’s home) is systematics + statistics. Branch 2 comprises **quintom** (two fields [14]), **modified gravity**, **interacting dark sector**, **early-universe/sound-horizon**, and **holographic** models.

Acknowledged asymmetry (scope limitation). The systematics branch received leave-one-out and cross-probe stress tests; the real-physics branches are summarized but not subjected to the same quantitative scrutiny here. Their broad disfavoring reflects the cited literature, not equivalent independent testing in this work — a deliberate scope boundary, not a verdict against them.

6 Method lesson (transferable)

Coordinate choice can flip a residual-analysis conclusion. The *same* LRG residuals told a “coherent line-of-sight systematic” story in the $(D_M/r_d, D_H/r_d)$ basis and a “two independent fluctuations” story in the physically natural $(D_V/r_d, F_{AP})$ Alcock–Paczynski basis. Rotate into the basis where the physically meaningful combinations are the axes *before* judging coherence.

References

- [1] B. Popovic, P. Shah, W. D. Kenworthy, et al., *The Dark Energy Survey Supernova Program: A Re-analysis of Cosmology Results and Evidence for Evolving Dark Energy with an Updated Type Ia Supernova Calibration*, MNRAS **548**, 4 (2026), arXiv:2511.07517. [DES-Dovekie: $\Omega_m = 0.330 \pm 0.015$; $w_0 = -0.803 \pm 0.054$, $w_a = -0.72 \pm 0.21$; significance reduced $4.2\sigma \rightarrow 3.2\sigma$.]
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- [3] L. Huang, R.-G. Cai & S.-J. Wang, *The DESI DR1/DR2 Evidence for Dynamical Dark Energy is Biased*

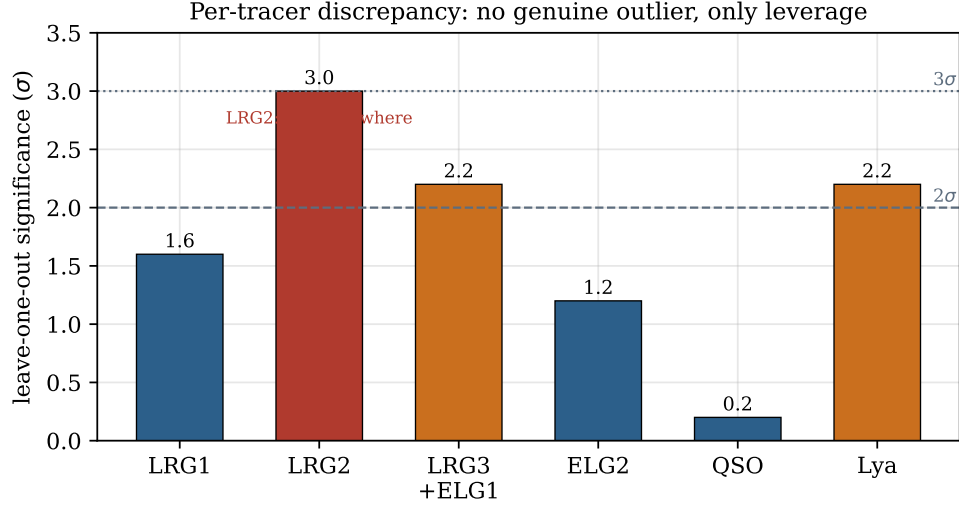


Figure 2: Leave-one-out significance per tracer (fit Λ CDM + CMB to all others, measure the held-out tracer’s joint $(D_M/r_d, D_H/r_d)$ pull). LRG1 is only 1.6σ ; LRG2 drives the effect at 3.0σ ($\sim 2.4\sigma$ after look-elsewhere). No genuine outlier — the signal is leverage, not anomaly. (BGS omitted: single observable.)

by *Low-Redshift Supernovae*, Sci. China Phys. Mech. Astron. **68**, 100413 (2025), arXiv:2502.04212. [The a_B intercept diagnostic; correcting the low- z systematic reduces the preference to $< 2\sigma$.]

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- [6] DESI Collaboration, *DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints*, arXiv:2503.14738 (2025).
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- [9] *Comparing Λ CDM, w CDM, and w_0w_a CDM Models with DESI DR2 BAO: Redshift-Resolved Diagnostics and the Role of r_d* , arXiv:2507.01380 (2025). [Block-diagonal covariance, $(D_V/r_d, D_M/D_H)$ and $(D_M/r_d, D_H/r_d)$ intra-bin blocks.]
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- [11] S. Afroz & S. Mukherjee, *Hint Towards Inconsistency Between BAO and Supernovae Dataset: The Evidence of Redshift-Evolving Dark Energy from DESI DR2 is Absent*, arXiv:2504.16868 (2025).

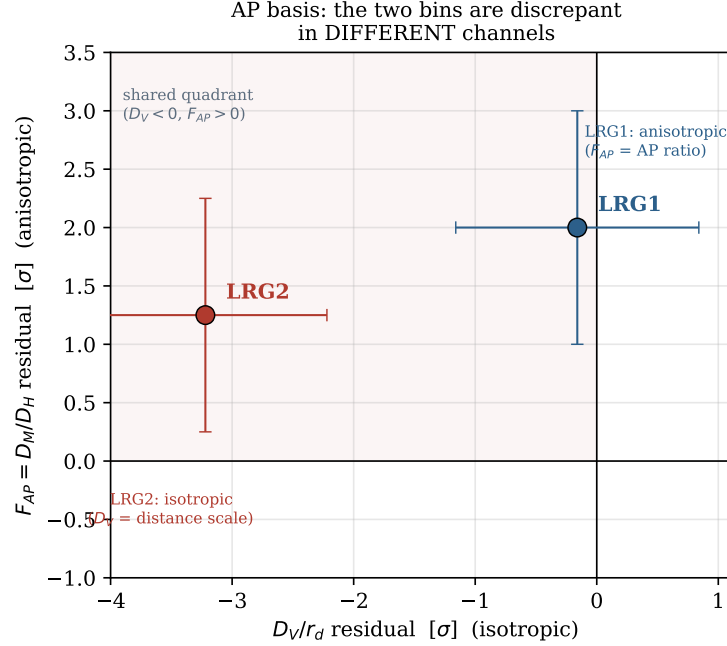


Figure 3: The LRG residuals in the Alcock–Paczynski basis. LRG1’s discrepancy is almost entirely anisotropic (F_{AP}); LRG2’s is almost entirely isotropic (D_V/r_d). They share quadrant but not shape — inconsistent with a single shared systematic or a single real feature, and more consistent with independent scatter. Error bars are 1σ ; LRG2’s D_V/r_d magnitude is pipeline-conditional (§1.2).

- [12] L. Perivolaropoulos, *Vacuum Energy, the Cosmological Constant, and UV Cutoffs*, arXiv:0802.1531 (2008).
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- [14] Y.-F. Cai, X. Ren, T. Qiu, M. Li & X. Zhang, *The Quintom Theory of Dark Energy After DESI DR2*, Natl. Sci. Rev. (2026), arXiv:2505.24732.

Note on numbers. Several absolute significances come from a simplified pipeline (compressed CMB distance priors, magnitude proxies) that inflates absolute values; the robust content is in directions, relative shifts, and reproduced structural results. Where a claim rests on the pipeline’s absolute calibration it is flagged as method-conditional, and the corresponding full-likelihood computation is listed as open work in §4.

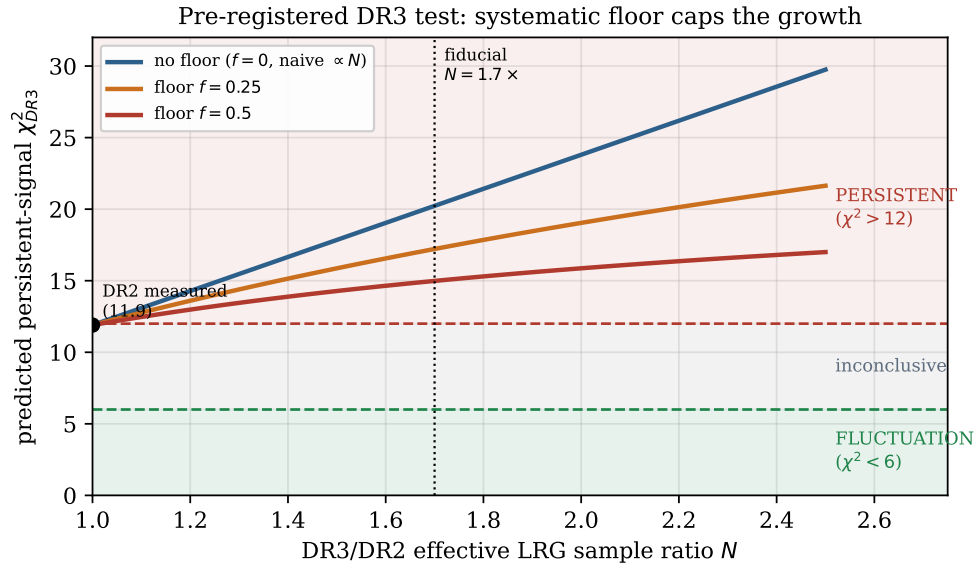


Figure 4: The pre-registered DR3 test with a systematic-error floor. Naive $1/\sqrt{N}$ scaling (blue) predicts a persistent signal reaches 4.1σ ; allowing a systematic floor ($f = 0.25, 0.5$) softens this to $3.5\text{--}3.7\sigma$ and widens the inconclusive band. The fluctuation hypothesis instead regresses toward $\chi^2 \approx 2$. The decision boundary ($\chi^2 < 6$ vs > 12) is robust across the floor range.